Efficient Data Management and Organization

Project Overview:

The Data Management and Analysis Project aimed to streamline the process of handling a large, diverse dataset containing images of various species. The primary focus was on optimizing the process of downloading, organizing, accuracy of data handling and analyzing the dataset for various experiments. This initiative aimed to address challenges such as data inconsistencies, enhance data download efficiency, and minimize manual classification errors, sought to enhance the overall quality and accessibility of the dataset for future research and analysis.

Project Phases:

1. **Efficient Data Download and Organization:**

➢ *Challenge*: Inconsistencies and loss during download.

➢ *Resolution:*

- Implemented an efficient data download process with pagination to avoid data loss.
- Standardized species names to prevent duplication and misspellings.
- Transformed data into species-wise format for easier analysis.
- Proposed a manual data handling approach for updates.
- Organized data in day-wise and species-wise formats for accessibility.

2. **Exploratory Data Analysis (EDA):**

➢ Created visualizations for dataset analysis in the jupyter notebook.

➢ Developed comprehensive dataframes for day-wise and species-wise data.

➢ The notebook generates an [Excel file] with all the details about the dataset till the last date the dataset was updated.

➢ Prepared detailed documentation for each code file for clarity and ease of use.

3. Dataset Growth:

➢ Successfully increased the dataset size to 16,000 images, up from 2,500 from May, June and July with pagination implemented in August.

4. Reasoning for Classification:

➢ Developed a code file for automatic organization based on given classification reasons.

➢ Enhanced efficiency in data management and improved transparency.

5. App Testing Data Handling:

➢ Implemented a code file for downloading and organizing app testing data.

6. Misclassified Images Handling:

➢ Identifying and managing misclassified images.

➢ Established a misclassified folder with an automated correction process.

➢ Updated data collection files to remove existing misclassifications.

➢ Modified dataset EDA file for misclassified data analysis.

7. Species Classification Project:

➢ Working on species classification model to prevent misclassifications.

8. **Clear and Blur Classification Project:**

➢ Planning to work on a clear and blur classification project to further enhance dataset quality.

9. **Single and Group Classification Project:**

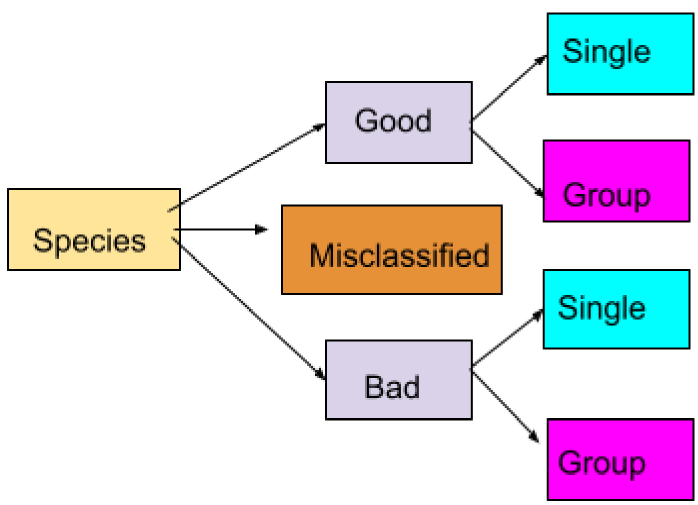
➢ Developed a common model for fish, prawns, and crabs for both single and group classification.

➢ Also developed individual models for fish and prawns.

➢ No separate model for crabs due to data insufficiency.

➢ Updated code for data collection and dataset EDA to accommodate single and group classification.

➢ Modified dataset organization to include categories for single and group images.



Challenges and Lessons Learned:

➢ Addressed challenges related to data inconsistencies and loss.

➢ Recognized the importance of ongoing manual checks for data quality.

➢ Adaptability was crucial in responding to evolving project requirements.

Conclusion:

The Data Management and organization Project has significantly improved the efficiency and quality of data handling. The implemented solutions have laid the groundwork for future projects, and ongoing efforts continue to enhance the dataset for more accurate and reliable analyses.